

SS2P2

Stable State-Space Point Process

$SS2P2 = S2P2 \text{ LLH backbone (expressive, parallel-scan)} \rightarrow \text{gated-bounded (G1) rate}$ $\lambda(t) \times \text{rate-neutral mark head } p^*(x t).$

Three pieces.

1. **Backbone (keep S2P2’s expressivity).** Stacked latent-linear-Hawkes layers with complex diagonal modes, ZOH-discretized and *parallel-scan* computable, produce a history embedding

$$u(t) \in \mathbb{R}^H \quad (\text{the same deep, expressive state S2P2 learns}).$$

2. **Rate head (make it simulation-stable).** A gated bound (G1) maps u to a *two-sided bounded* total rate:

$$o = \sigma(W_o u + b_o) \in (0, 1)^H, \quad h = o \odot \tanh(u) \in (-1, 1)^H, \quad \lambda(t) = \text{softplus}(w^\top h + b_\lambda).$$

Since $|h_i| < 1$, $\lambda(t) \in (\ell_-, \ell_+)$ with $\ell_\pm = \text{softplus}(b_\lambda \pm \|w\|_1)$ — a “Poisson sandwich” $\text{Poisson}(\ell_-) \preceq N(t) \preceq \text{Poisson}(\ell_+)$: *no runaway, no death*, stable *without* a branching projection.

3. **Mark head (rate-neutral / decoupled).** Marks read the *same* u but do not change the total rate:

$$\text{single-item: } p^*(\{k\} | t) = \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad \text{set: } \rho_k = \sigma(z_k), \quad z = Vu + c, \quad \sum_x p^*(x | t) = 1,$$

$\lambda_x^*(t) = \lambda(t) p^*(x t)$	$\left(\sum_x \lambda_x^* = \lambda : \text{marks never perturb the bound} \right).$
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Why it works. The backbone keeps S2P2’s representational power; the G1 gate makes the closed-loop rate *provably bounded* (so free-running simulation can’t blow up or die); decoupling lets the expressive marks ride on a stable rate. Calibration is optional — the bias b_λ and the MLE $-\int \lambda$ term set the mean; the bound holds regardless — so SS2P2 needs *no* EMA and *no* mean-pin to stay stable.